

FAQ Assistant API

Project Testing Phase Document

Item	Details
Project Name	FAQ Assistant API
Document	Testing Phase (Test Plan)
Version	1.0 (Draft)
Phase	Phase 5 - Testing
Status	For review

1. Introduction

This document explains how the FAQ Assistant API will be tested after development. It defines the test objectives, scope, strategy, environment, test cases, defect handling and the criteria used to decide whether the API is ready for deployment.

2. Testing Objectives

- Verify that every feature works as described in the requirements.
- Confirm that the API returns correct and relevant answers with a proper confidence score.
- Find and fix defects before the API is released.
- Check security, performance and error handling under normal and abnormal use.
- Provide a clear test report to decide release readiness.

3. Scope of Testing

3.1 In Scope

- All endpoints: /ask, /faqs, /feedback, /unanswered and /health.
- Matching engine accuracy and fallback behaviour.
- Authentication, authorization and input validation.
- Database operations, logging and feedback storage.
- Response time, load handling and rate limiting.
- API documentation (Swagger) accuracy.

3.2 Out of Scope

- Front-end or client application testing.
- Third-party hosting and network infrastructure testing.
- Voice, multi-language and live-agent features (not part of this version).

4. Test Strategy

Testing is done in levels, starting from small units of code and moving up to the complete system.

Level	What is Tested	Done By	Tools
Unit Testing	Individual functions such as text cleaning, scoring, validators	Developers	Pytest
Integration Testing	Modules working together (API, database, matcher)	Developers / QA	Pytest, Test DB
System / API Testing	Complete endpoints against requirements	QA	Postman, Swagger
Security Testing	Auth, injection, access control	QA	Postman, OWASP ZAP

Level	What is Tested	Done By	Tools
Performance Testing	Speed and stability under load	QA	Locust / JMeter
Regression Testing	Old features still work after fixes	QA	Automated suite
User Acceptance Testing	Owner confirms the API meets needs	Project Owner	Postman collection

5. Test Environment

Item	Details
Environment	Separate test server, isolated from production
Database	PostgreSQL test database with sample FAQ data
Runtime	Same language version and Docker image as deployment
Tools	Pytest, Postman, Locust, OWASP ZAP, coverage report
Test Data	Set of FAQs, reworded questions, invalid inputs and unrelated questions

6. Entry and Exit Criteria

6.1 Entry Criteria

- Development phase is complete and the build is deployed to the test environment.
- Requirements, design and API documents are available.
- Test data and test cases are prepared and reviewed.
- Unit tests written by developers are passing.

6.2 Exit Criteria

- All planned test cases are executed.
- 100 percent of critical and high priority test cases pass.
- No open critical or high severity defects.
- Matching accuracy meets the agreed target on the test question set.
- Test summary report is approved by the project owner.

7. Functional Test Cases

ID	Test Case	Input / Steps	Expected Result
TC01	Ask exact question	POST /ask with a question stored in FAQ	200 OK, correct answer, confidence above threshold
TC02	Ask reworded question	POST /ask with same meaning, different words	Correct FAQ returned with good score

ID	Test Case	Input / Steps	Expected Result
TC03	Ask unrelated question	POST /ask with a question not in FAQ	Fallback message, question saved as unanswered
TC04	Empty question	POST /ask with empty text	400 Bad Request with clear message
TC05	Very long question	POST /ask with text above allowed limit	400 Bad Request
TC06	Create FAQ (admin)	POST /faqs with valid token and data	201 Created, FAQ saved
TC07	Create FAQ without token	POST /faqs with no token	401 Unauthorized
TC08	Update FAQ	PUT /faqs/{id} with new answer	200 OK, updated answer returned by /ask
TC09	Delete FAQ	DELETE /faqs/{id}	200 OK, FAQ no longer returned
TC10	Get missing FAQ	GET /faqs/9999	404 Not Found
TC11	Submit feedback	POST /feedback with helpful flag	201 Created, feedback stored
TC12	Health check	GET /health	200 OK, status running

8. Matching Accuracy Testing

Because the main value of the API is giving the right answer, a separate accuracy test is done.

- Prepare a test set of at least 50 questions: exact, reworded, misspelled and unrelated.
- Record for each question the expected FAQ (or none for unrelated questions).
- Run all questions through /ask and compare the returned FAQ with the expected one.
- Calculate accuracy = correct answers / total questions, and check fallback for unrelated questions.
- If accuracy is below target, tune the threshold or matching method and test again.

Question Type	Example	Expected
Exact	How do I reset my password?	Matching FAQ
Reworded	I forgot my password, what now?	Matching FAQ
Misspelled	how to reest pasword	Matching FAQ
Unrelated	What is the weather today?	Fallback reply

9. Security Testing

Check	Method	Expected Result
Authentication	Call admin endpoints with no, wrong and expired tokens	Access denied (401 / 403)

Check	Method	Expected Result
Authorization	Use a normal key on admin-only endpoints	Access denied (403)
Injection	Send SQL and script text in inputs	Input safely handled, no error or data leak
Data exposure	Check responses and error messages	No secrets, keys or stack traces shown
Rate limiting	Send many requests quickly	429 Too Many Requests after the limit

10. Performance Testing

Test	Description	Target
Response time	Average time for /ask under normal load	Under 500 ms
Load test	Simulate expected number of concurrent users	No failures, stable time
Stress test	Increase load beyond the expected limit	Fails gracefully, recovers after
Endurance test	Run moderate load for a long period	No memory leak or slowdown

11. Defect Management

Every defect is logged with a title, steps to reproduce, expected and actual result, severity and screenshots or logs. Defects go through this life cycle: New, Assigned, In Progress, Fixed, Retest, Closed (or Reopened if the problem remains).

Severity	Meaning	Example	Fix Time
Critical	System unusable or data loss	API crashes on every /ask call	Immediately
High	Major feature broken, no workaround	Admin cannot create FAQs	Within 1 day
Medium	Feature works partly, workaround exists	Wrong score shown for some inputs	Within 3 days
Low	Minor or cosmetic issue	Typo in error message or docs	Next release

12. Testing Schedule

Activity	Duration	Output
Test planning and case preparation	2 days	Test plan and test cases
Integration and API testing	3 days	Execution results, defect list
Security and performance testing	2 days	Security and performance reports
Bug fixing and retesting	3 days	Fixed and closed defects
Regression testing	1 day	Regression results
User acceptance testing and sign-off	1-2 days	Test summary report, sign-off

13. Roles and Responsibilities

Role	Responsibility
Test Lead / QA	Prepare plan, assign work, track defects, write the report
Testers	Design and execute test cases, log defects, retest fixes
Developers	Fix defects, support unit and integration tests
Project Owner	Perform acceptance testing and approve release

14. Risks and Mitigation

Risk	Mitigation
Not enough real questions for accuracy testing	Collect questions from support team or users early
Late defects delay release	Test early, run automated tests on every build
Test environment differs from production	Use the same Docker image and configuration
Too many defects in one area	Extra review and regression testing of that module

15. Deliverables

- Test Plan (this document) and detailed test case sheet.
- Test execution report with pass and fail status.
- Defect report with severity and status of every bug.
- Accuracy, security and performance test reports.
- Test summary report with release recommendation.

16. Next Phase

After the exit criteria are met and the report is approved, the project moves to Phase 6 - Deployment, where the API is released and the user guide is prepared.